

PROFESSIONAL SUMMARY

Embedded instrumentation systems integrator with 10+ years of experience in electronics and software design. Executing all the embedded system developing stages: system development from conception, firmware design, programming and testing, hardware prototyping, and final PCB design and testing. Experienced in bioelectronics project development, general instrumentation design, and testing.

SKILLS

Soft Skills: | Problem Solving | Time Management | Team Work | Creativity | Adaptability |

Technical Skills: | Embedded C | C++ | Python | Embedded Linux | ASM | RTOS | Debugging | STM32 | ESP32 | ATmega | Microchip PIC | Raspberry Pi | BeagleBone Black | Arduino | Serial | I2C | SPI | One Wire | Timers | Git | STM32 Cube IDE | KiCad | Altium Designer | Visual Studio | Visual Studio Code | Visual Micro | MPLAB IDE | Troubleshooting | SMD Soldering | System Integration | Technologist | Analog Circuit Design | AD620 | INA114 | MAX4194 | MAX4094
| Mixed Signals Systems | Digital Multimeters | Oscilloscopes | Logic Analyzers |

PROFESSIONAL EXPERIENCE

Embedded systems developer, Moon Trades (Self-Employed Contractor), **September 2025 – Present**

- Leading sensor integration and embedded system architecture for the PEBBLE™ autonomous exploration platform.
- Selecting and integrating advanced instrumentation, including LiDAR, hyperspectral imaging, and XRF geochemical analysis systems.
- Developing a real-time processing architecture on NVIDIA Jetson Orin for GPS-denied navigation and edge AI, and executing hardware integration for OCI Discovery Project underground validation.

Embedded Systems Engineer, Robbie-Insurtech, (Self-Employed Contractor), **Aug 2025 – Present**

- Evaluate and document the new hardware sensor upgrade.
- Support software and hardware integration and development of hard-tech solutions.
- Validated hardware integration using Linux-based tools -Field-tested moisture sensors for performance and reliability.

Main Engineer, Rabbit Systems (Self-Employed Contractor), **Mar 2024 – Present**

- Designed a Command Line Interface library for STM32 and ESP32 microcontrollers.
- Designed a prototype of a CAS (Collision Avoidance System) for a racing car application.
- Designed and tested a low-power datalogger system for general-purpose applications.
- Execute requirements from verbal description to functional plan documentation, component selection, and layout and programming of a functional PCB using surface-mount soldering.

Embedded Systems Engineer, Morphace (Self-Employed Contractor), **October 2025 – April 2026**

- Designed embedded circuits and PCBs in KiCad for skin-health and bioelectronics applications
- Supported hardware validation and regulatory compliance under CSA SPE-3000:19

CARLOS NARVAEZ

EMBEDDED SYSTEMS DEVELOPER/ENGINEER

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Instrumentation & Embedded Systems Developer, Minard (Electronic Components Supplier)

Nov 2020 - Mar 2024

- Software to simulate a TRIAXYS wave sensor.
- Created a software architecture plan for a general-purpose command line interface for microcontrollers, which bound a serial ASCII command to microcontroller functions.
- Conducted systems and program architecture planning, schematic design, PCB layout, and firmware development, and integrated an OEM technology photoplethysmography sensor with a GUI using Java.
- Prototyped and developed a 3D-printed harmonic drive for low-cost robotics using a brushless motor.
- Developed a data logger system prototype for maritime deep sensing applications for the conductivity of seawater.
- Designed a wireless low-power electrocardiography data acquisition device to be used in laboratory rats.

Electronic Instrumentation Engineer, National University of Mexico, Feb 2018 - Oct 2020

- Led the process and collaborated with the mechanical and engineering teams to develop instrumentation devices for astronomical observations.
- Maintained robotic telescopes COATLI and DDOTI to the joint electronic system and the main computer.
- Produced reports and designed a testing device for astronomical CCD using ATMega 328 laser LED's.
- Designed a meteorological measurement system for reading weather conditions inside the main telescope dome using a commercial digital anemometer adapted to be read with a BeagleBone board.

Electrical & Electronic Engineering Professor, National Technology Institute of Mexico

Jan 2017 - Jan 2018

- Facilitated courses in Adv. Programming, Instrumentation RT Programming, and tutored on embedded sys.

LANGUAGES

Languages: | English (Professional Working Experience) | Spanish (Native Speaking) |

EDUCATION & CERTIFICATES

Computer Programmer Diploma, Seneca Polytechnic. 2023

Master's in Electronic Engineering, Autonomous University of San Luis Potosi. 2016

BEng. Mechatronics Engineering, Technological Institute of San Luis Potosi. 2013

INTERESTS

| Clay Sculpting | Urban Cycling | Homemade Electronic Projects | Hiking | Exploring the City |